

From: Doug Langworthy <dlangworthy@Cadillac-MI.net>
Sent: 11/28/2018 10:27:50 AM
To: "Sneller, Cindy (DEQ)" <SNELLERC@michigan.gov>
Subject: RE: Land appl docs and DTs submitted for Biosolids AR - Questions
Attachments: Class B Pathogen Reduction and Site Restriction Revised.pdf

Cindy,

I've attached the revised pathogen doc. I am contacting them again to see if they have the DT applied doc. As far as my figures on DT's I use our daily lab numbers which is a composite of every truck load hauled out for the day and the number of gallons hauled each day to figure my DT's. I had talked to Dave S. in the past and he said that was acceptable because of the discrepancies we've had with previous haulers numbers.

Thanks Doug

Douglas Langworthy

Assistant Water Resources Supervisor

City of Cadillac

231-775-2841

From: Sneller, Cindy (DEQ) [mailto:SNELLERC@michigan.gov]
Sent: Thursday, November 15, 2018 3:11 PM
To: Doug Langworthy
Subject: Land appl docs and DTs submitted for Biosolids AR - Questions
Importance: High

Hi Doug – I'm trying to figure out the land application documents you submitted for the Annual Report. One is in an Excel spreadsheet and the others are pdfs of Ag rate calculations.

I cannot make heads or tails out of the Ag rate calculations. I'm trying to find on these sheets **how many Dry Tons (DT) of Biosolids were land applied on these fields** and I can't find that anywhere.

The one site you uploaded as an *Excel spreadsheet*, is nice because I can find the DT information.

Do you have sheets for each one of the other sites with the same info as the one (2018 As applied-29NS01)? Or, at least something for the other sites that tells me how many DTs of biosolids were applied to the site?

The other information is fine to submit, it's just that not all the information is on these sheets.

Also, the Applier from Iowa for this past Spring, on their certification form for Pathogen and Site Restrictions, they marked – 7 day fecal and Geometric mean. Cadillac doesn't do that one, you guys do Anaerobic Digestion. See my attachment, the process they should've marked.

To finish up this report review, I will need the following:

- A revised Pathogen & Site Restriction form from the Applier for the Spring haul.
- Documentation showing how many DTs per site were applied.

Then I can correctly figure out the DTs per your facility for the year. **Right now, I calculated 274 DTs and you reported 240.87 DTs.** That's a big difference.

Let me know if you have questions. I'm only in until 3:30 pm tomorrow (Fri. Nov. 16) and will be on vacation after that until Dec. 5, returning on that day at 7 am.

Cindy

Cindy A. Sneller, Senior Environmental Quality Analyst
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ATTACHMENT NAME:

Class B Pathogen Reduction and Site Restriction Revised.pdf

ATTACHMENT TYPE:

Adobe Portable Document Format (PDF) compound image

FACILITY: Cadillac WWTFCITY: Cadillac MI**CLASS B PATHOGEN REDUCTION AND SITE RESTRICTIONS**

How were the class B pathogen reduction and site restriction requirements met?

Site Identification:

Are the site identification for this form the same as previously identified on the site description form?

☒ Yes ☐ No If no, provide a correct site description form for each site and attach it to this form.

CHECK ALL THAT APPLY

- 1) ☐ The geometric mean of the density of fecal coliform in the samples is less than 2,000,000 Most Probable Number (MPN) / Colony Forming Units (CFU) per gram of total solids (dry weight basis).
Fecal Coliform geometric mean was _____ MPN / CFU per dry gram total solids for the reporting period.

Go to question 4

- 2) ☒ A Process to Significantly Reduce Pathogens (PSRP) was used to meet Class B pathogen reduction.

CHECK ALL PSRP USED TO MEET THE CLASS B PATHOGEN REDUCTION REQUIREMENT.

☐ Aerobic Digestion

Aerobic conditions were maintained for a mean cell residence time and temperature between 40 days at 20 C (68 F) or 60 days at 15 C (59 F).

☐ Air drying

Sewage sludge was dried on sand beds or on paved basins. The sewage sludge dried for a minimum of three (3) months. During two (2) of the three (3) months, the ambient average daily temperature was above 0 C (32 F).

☒ Anaerobic Digestion

Sewage sludge was treated in the absence of air for a specific mean cell residence time and temperature between 15 days at 35 to 55 C (95 to 131 F) or 60 days at 20 C (68 F).

☐ Composting

Using either the within-vessel, static aerated pile, or windrow composting methods, the temperature of the sewage sludge was raised to 40 C (104 F) or higher and remained at 40 C (104 F) or higher for five (5) days. For four (4) hours during the five (5) days, the temperature in the compost pile exceeded 55 C (131 F).

☐ Lime Stabilization

Sufficient lime was added to the sewage sludge to raise the pH of the sewage sludge to 12 after two (2) hours of contact.

- 3) ☐ An Equivalent PSRP was used to meet Class B pathogen reduction.

☐ EPA Equivalency Statement is included.☐ EPA Equivalency Statement was provided previously.

CLASS B SITE RESTRICTIONS

- 4) Are food crops grown on the site receiving Class B sludge?
(Food crops are human food normally eaten raw or with little processing.)

Site	All	1	2	3	4	5	6	7	8	9	10
No	_____	<u>x</u>	<u>x</u>	<u>x</u>	<u>x</u>	_____	_____	_____	_____	_____	_____

If no to all sites, go to question 6.

Yes _____

- 4A) Do harvested parts above the surface of the land touch the soil/sludge mixture?

Site	All	1	2	3	4	5	6	7	8	9	10
No	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____
Yes	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____

If **no** to all sites, go to question 4C.

If **yes** to any site, read the following and answer question 4B.

Food crops with harvested parts that touch the sewage sludge/soil mixture and are totally above the land surface shall not be harvested for 14 months after application of sewage sludge.

- 4B) Is food harvest prohibited for 14 months or more after the last sludge application?

Site	All	1	2	3	4	5	6	7	8	9	10
No	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____
Yes	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____

- 4C) Are harvested parts below the surface of the land?

Site	All	1	2	3	4	5	6	7	8	9	10
No	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____
Yes	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____

If **no** to all sites, go to question 5

If **yes** to any site, read the following and answer question 4D.

Food crops with harvested parts below the surface of the land shall not be harvested for 20 months after application of sewage sludge when the sewage sludge remains on the land surface for four (4) months or longer prior to incorporation into the soil.

- 4D) Does the sewage sludge remain on the land surface for four (4) months or longer prior to the incorporation into the soil?

Site	All	1	2	3	4	5	6	7	8	9	10
No	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____
Yes	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____

If **no** to all sites, go to question 4F.

If **yes** to any site, read the following and answer question 4E.

4E) Is food harvest prohibited for 20 months or more after the last sludge application?

Site	All	1	2	3	4	5	6	7	8	9	10
No	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____
Yes	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____

Go to question 5.

4F) Food crops with harvested parts below the surface of the land shall not be harvested for 38 months after application of sewage sludge when the sewage sludge remains on the land surface for less than four (4) months prior to incorporation into the soil.

Is food harvested prohibited for 38 months or more after the last sludge application?

Site	All	1	2	3	4	5	6	7	8	9	10
No	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____
Yes	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____

Go to question 5.

5) Food crops shall not be harvested for 30 days after application of sewage sludge.

Is all food crop harvesting prohibited for 30 days or more after the last sludge application?

Site	All	1	2	3	4	5	6	7	8	9	10
No	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____
Yes	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____

6) Are animal feed (such as corn, milo, wheat, soybeans, grasses, hay, alfalfa, etc.) or fiber crops grown on the site receiving Class B sludge?

Site	All	1	2	3	4	5	6	7	8	9	10
No	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____
Yes	_____	<u>x</u>	<u>x</u>	<u>x</u>	<u>x</u>	_____	_____	_____	_____	_____	_____

If **no** to all sites, go to question 7.

If **yes** to any site, read the following and answer question 6A.

Feed crops and fiber crops shall not be harvested for 30 days after application of sewage sludge.

6A) Are harvest of feed and fiber crops prohibited for 30 days or more after the last sludge application?

Site	All	1	2	3	4	5	6	7	8	9	10
No	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____
Yes	_____	<u>x</u>	<u>x</u>	<u>x</u>	<u>x</u>	_____	_____	_____	_____	_____	_____

Go to question 7.

7) Are animals allowed to graze on the site receiving Class B sludge?

Site	All	1	2	3	4	5	6	7	8	9	10
No	_____	<u>x</u>	<u>x</u>	<u>x</u>	<u>x</u>	_____	_____	_____	_____	_____	_____
Yes	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____

If **no** to all sites, go to question 8.

If **yes** to any site, read the following and answer question 7A.

Animals shall not be allowed to graze on the land for 30 days after application of sewage sludge.

7A) Are animals prohibited from grazing on the land for 30 days after the last sludge application?

Site	All	1	2	3	4	5	6	7	8	9	10
No	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____
Yes	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____

Go to question 8.

8) Is turf grown on the site receiving Class B sludge?

Site	All	1	2	3	4	5	6	7	8	9	10
No	_____	<u> x </u>	<u> x </u>	<u> x </u>	<u> x </u>	_____	_____	_____	_____	_____	_____
Yes	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____

If **no** to all sites, go to question 9.

If **yes** to any site, read the following and answer question 8A.

Turf grown on land where sewage sludge is applied shall not be harvested for one (1) year after application of the sewage sludge when the harvested turf is placed on either land with a high potential for public exposure or a lawn, unless otherwise specified by the permitting authority. A high potential for public exposure area is a private or public lawn, cemetery, ball field, playground, park, golf courses, etc...

8A) Was turf harvest prohibited for one (1) year after the last sludge application date?

Site	All	1	2	3	4	5	6	7	8	9	10
No	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____
Yes	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____

Go to question 9.

9) Public access to land with a high potential for public exposure shall be restricted for one (1) year after application of sewage sludge. Was this requirement met

Site	All	1	2	3	4	5	6	7	8	9	10
No	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____
Yes	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____
N/A*	_____	<u> x </u>	<u> x </u>	<u> x </u>	<u> x </u>	_____	_____	_____	_____	_____	_____

*Not Applicable

Go to question 10.

10) Public access to land with a low potential for public exposure shall be restricted for 30 days after application of sewage sludge. Was this requirement met?

Site	All	1	2	3	4	5	6	7	8	9	10
No	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____
Yes	_____	<u> x </u>	<u> x </u>	<u> x </u>	<u> x </u>	_____	_____	_____	_____	_____	_____
N/A*	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____	_____

*Not Applicable

Go to question 11.

11) Indicate the method of restricting public access.

CHECK ALL THAT APPLY.

☐ Information Signs

☒ Fenced

☒ Rural Area / posted at entrances

☒ Rural Remote

☐ Other _____

CERTIFICATION

I certify under the penalty of law that the information as listed above is complete and accurate to the best of my knowledge. This determination has been made under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information used to determine the management practices have been met. I am aware that there are significant penalties for false certification, including the possibility of fine and imprisonment.

Name and Official Title (type or print)

Signature

Date Signed

Rev. 4/00

FORM LA-PR